

want to define an LLM version of the above n -gram conditional distribution using the LLM predicted probability of s^y in the context of the concatenated testing example $u \oplus x \oplus y$:

$$P_{\text{LLM}}(s^y|s^x) = \prod_{t \in s^y} P_{\text{LLM}}(t|u \oplus x \oplus y_{[1:m-1]}). \quad (2)$$

Here m denotes the location index of the n -gram s^y found in y , and t is each token in the tokenized n -gram s^y . We can then formally define the **distributional memorization** by Spearman correlation ρ between the task-gram language model probabilities and LLM predicted probabilities of testing data:

Definition 3. For a testing set $D'_T = \{(x_i, y_i)\}_i$, we denote all n -gram pairs found in it by $\Phi = \{(s^x, s^y) | \forall (s^x, s^y) \in [G_n(x) \times G_n(y)] \cap H_n(T), \forall (x, y) \in D'_T\}$. Then we define the extent of an LLM **distributional memorize** the pretraining corpus $\mathcal{D}$ when performing task T as follows:

$$\text{Mem}_n(\text{LLM}, \mathcal{D}|T) = \rho(\log P_{n,\mathcal{D}}(Y|X), \log P_{\text{LLM}}(Y|X)), \quad (3)$$

where ρ denotes Spearman correlation, $\log P_{n,\mathcal{D}}(Y|X) = \{\log P_{n,\mathcal{D}}(s^y|s^x) | \forall (s^x, s^y) \in \Phi\}$, and $\log P_{\text{LLM}}(Y|X) = \{\log P_{\text{LLM}}(s^y|s^x) | \forall (s^x, s^y) \in \Phi\}$.

Similar to normal Spearman correlations, the significance of distributional generalization can be measured by p -value. The **distributional generalization** is then defined as the opposite of the distributional memorization: increased memorization implies decreased generalization, as the LLM predictions and the n -gram LM are more distributionally correlated, and vice versa.

Distributional memorization describes to what extent the LLM-predicted probability of the ground truth testing data can be viewed as a monotonic function of task-gram LM probability. This can be viewed as a measure of the predictability of task-gram LM to the LLM probability. In the following sections, we show comprehensive empirical evidence of LLMs performing knowledge-intensive tasks depending on distributional memorization while performing reasoning-intensive tasks depending on distributional generalization, as shown in Figure 2.

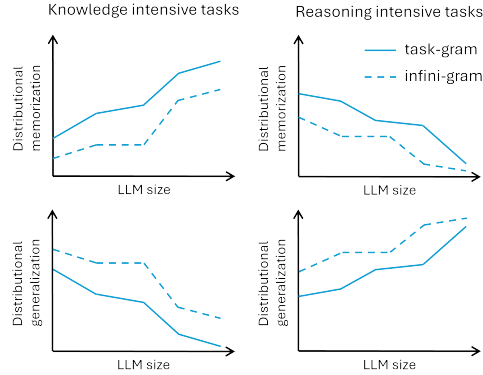


Figure 2: Expected distributional memorization and generalization trend for different types of tasks.

3 EXPERIMENTAL SETUP

In this section, we introduce the datasets, models, and tools we use for analyzing the memorization and generalization behaviors of LLMs.

Models and Pretraining Corpus We use a family of fully open-sourced, Transformer-decoder-based LMs: Pythia (Biderman et al., 2023), with a wide range of model sizes ranging from 13M to 12B parameters. All Pythia models are trained on Pile (Gao et al., 2020), a diverse pretraining corpus consisting of approximately 207B tokens. We also include some results with 1B and 7B OLMo models (Groeneveld et al., 2024), pretrained on Dolma (Soldaini et al., 2024b) with 3T tokens.

Downstream Tasks We use four types of tasks: machine translation, factual question answering, world knowledge questions, and reasoning.

For translation, we use the **WMT-09** dataset (Callison-Burch et al., 2009) with a 2.5K testing set. WMT is a classic annual machine translation shared task with different languages. We chose WMT09 as our testing data instead of newer versions of WMT because WMT09 contains more European languages, which is more prominent in the Pile.

For factual question answering, we use the **TriviaQA** dataset (Joshi et al., 2017) with a 10K testing set, which is a knowledge-intensive question-answering dataset with the questions originating from trivia enthusiasts. Since the answers are usually single words or short phrases, we regard the whole answer text as the output n -gram s^y no matter the value of n .

For world knowledge questions, we use the **MMLU** benchmark (Hendrycks et al., 2020), covering 57 tasks including elementary mathematics, US history, computer science, law, and more. The aim of

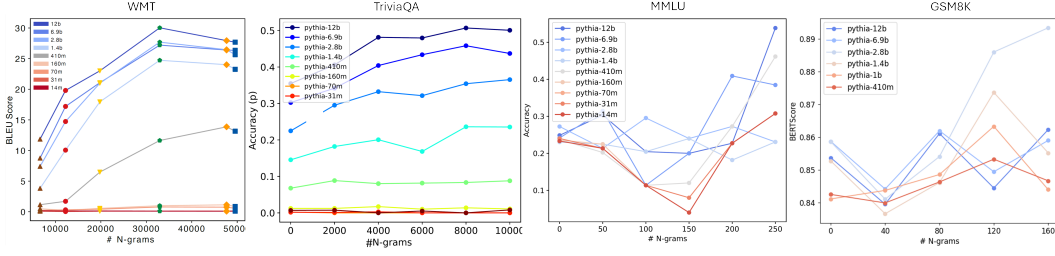


Figure 3: Task performance v.s. n -gram pair count in the Pile with different Pythia model sizes, for four different tasks, from left to right: WMT, TriviaQA, MMLU, and GSM8K.

For WMT, the x -axis shows the counts of n -gram pairs found in the testing set of six different languages: Hungarian, Czech, German, Italian, Spanish, and French, from left to right.

MMLU is to test models’ world knowledge and problem-solving ability, so different from TriviaQA, there is a large portion of questions that require logical/math reasoning skills.

For complex reasoning, we use the **GSM8K** dataset (Cobbe et al., 2021), which contains linguistically diverse grade school math world problems with step-by-step solutions. To solve GSM8K, the LLM needs to first generate the chain-of-thoughts (CoT) reasoning step, and then draw the final conclusion, which requires advance math and logical reasoning capabilities.

Searching over Pretraining Data at Scale Given the scale of the LLM pretraining corpus, searching n -grams over the whole corpus $\mathcal{D}$ is non-trivial. We use *What’s In My Big Data?* (**WIMBD**) (Elazar et al., 2024) and the ∞ -gram (Liu et al., 2024) platform, which are designed to search and retrieve documents in huge corpora. Both of them have indexed the Pile, allowing us to search it using API calls. We use WIMBD for accurately counting the co-occurrence of n -gram pairs as the co-occurrence frequency is usually low, and the approximate counting function in ∞ -gram sometimes fails to capture these low-frequency n -gram pairs. We use ∞ -gram for counting the occurrence of single n -grams as they appear more frequently in the corpus and the counting approximation has a relatively small effect on the search results. We also use the ∞ -gram API to produce the ∞ -gram probability of single n -grams, as detailed in Section 5.

4 n -GRAM DATA FREQUENCY V.S. TASK PERFORMANCE

In this section, we investigate the overall relation between the n -gram pair counts and the LLM task performance, which can be viewed as a rough estimation of the importance of the task-related data. We confirm that the Pile is not contaminated by any of the datasets we used, by ensuring there are no large n -grams ($n = 8$ and $n = 14$) overlaps between the Pile and the testing data. This decontamination method is adopted from the GPT3 technical report (Brown, 2020).

We estimate the probability of a test example (x, y) appearing in the pretraining corpus as the probability of any of the n -gram pairs from the task-gram table found in (x, y) appearing:

$$P_{\mathcal{D},n}(x, y) \propto \sum_{(s^x, s^y) \in (x, y)} C((s^x, s^y), \mathcal{D}) \mathbb{1}_{(s^x, s^y) \in H_n(T)} \quad (4)$$

In Figure 3, we plot the task performances v.s. the count of n -gram pairs per example as defined in Equation (4), for WMT ($n = 2$), TriviaQA ($n = 5$), and MMLU ($n = 3$). For WMT, since the test set size for each language is the same, thus we show the sum of count per language here. For **TriviaQA**, **MMLU**, and **GSM8K**, the x -axis represents bins that group test examples based on the number of n -gram pairs. For instance, in the case of TriviaQA, data points at $x = 0$ correspond to test examples where the number of n -gram pairs falls between 0 and 2000. In general, TriviaQA has more n -gram pair counts than WMT, MMLU, and GSM8K.

The **WMT** performance is evaluated by the BLEU score between greedily generated translation and the reference translations. The **TriviaQA** performance is evaluated by the accuracy of the exact match of the generated answer. The **MMLU** performance is evaluated by the accuracy of the option

with the highest LM predicted probability. Since there are four choices for each question, the random performance of MMLU is 25%. For **GSM8K**, since the performance of Pythia models is low ($< 5\%$ accuracy), the accuracy plot is too sparse to show any trend. We compute the BERTScore (Zhang* et al., 2020) (precision) between the model-generated chain-of-thoughts (CoT) and the ground truth CoT instead.

When the model size is small ($< 410\text{m}$), **WMT** and **TriviaQA** have near-zero performance regardless of the n -gram pair count, while interestingly, **MMLU** reaches the lowest performance ($< 10\%$) significantly lower than random guessing when the n -gram pair count is around 150. A closer inspection of the test examples in this interval reveals that they contain more reasoning or math problems, which appear to be harder for Pythia models. The noisier performance curve of MMLU and GSM8K is likely due to the weaker capabilities of Pythia models on this benchmark.

In general, all task performance increases when the number of task-related n -gram pairs increases when the model size is large enough ($> 410\text{m}$). And the trend of performance improvement is more significant when the model size is larger. For GSM8K, the Pythia 2.8B model shows the most significant increasing trend, while larger models show a less significant trend. This seems to indicate that memorization plays an essential role in LM’s capabilities for all four tasks, and larger models memorize more. However, these performance curves can also be explained by the improved generalization ability of LMs when there is more relevant pretraining data. In the next section, we use pre-defined distributional memorization and generalization to investigate the possible causes behind these performance trends.

5 n -GRAM DISTRIBUTION V.S. LLM DISTRIBUTION

In this section, we compute the **distributional memorization** $\text{Mem}_n(\text{LLM}, \mathcal{D}|T)$ as defined in Definition 3, for $T = \text{WMT}, \text{TriviaQA}, \text{and MMLU}$ respectively. In addition to computing the distributional memorization with the **task-gram language model** as defined in Definition 2, we consider computing another version of **distributional memorization** with an n -gram language model defined by single n -grams. Specifically, we consider the ∞ -gram language model (Liu et al., 2024) that uses an n as large as possible for predicting the probability of each token, which is shown to be better aligned with human written text compared with classical n -gram LMs.

An ∞ -gram LM can be viewed as an n -gram LM initialized with $n = \infty$, and then backoff when the n -gram count equals zero. This way, the probability of each token is dependent on its longest prefix that exists in the pretraining corpus. Considering concatenating the input and output text $u \oplus x \oplus y$ as the context, such distribution can be written as

$$P_{\infty, \mathcal{D}}(s^y | u \oplus x \oplus y) = \prod_{t_i \in s^y} P_{\infty, \mathcal{D}}(t_i | t_{[1:i-1]}) = \prod_{t_i \in s^y} C(t_{[i-(n_i-1):i]}) / C(t_{[i-(n_i-1):i-1]}).$$

Here i is the location index of the token t_i in the concatenated text $u \oplus x \oplus y$, and n_i is the size of the longest prefix of t_i that can be found in the pretraining corpus, i.e., $n_i = \max \{n' \in [1, i] | C(t_{[i-(n'-1):i-1]}) > 0\}$. In practice, we ignore the tokens with zero probability and set the ∞ -gram probability of this token to one, i.e., $P_{\infty, \mathcal{D}}(s^y) = 0$ only when all its tokens have zero probability. Then the alternative version of **distributional memorization** using ∞ -gram LM is defined as:

$$\text{Mem}_{\infty}(\text{LLM}, \mathcal{D}|T) = \rho(\log P_{\infty, \mathcal{D}}(Y|X), \log P_{\text{LLM}}(Y|X)), \quad (5)$$

with all the notations similarly defined as in Definition 3. We then compute the two versions of distributional memorization for the different datasets, WMT, TriviaQA, and MMLU, and show the results in Figure 4.

Translation ability does not come from memorization. In Figure 4, we do not show any distributional memorization values of WMT because none of them are statistically significant. This indicates that while the translation performance is strongly positively correlated to the n -gram counts as shown in Figure 3, the performance gain does not come from initiating the pretraining data distribution.

To further investigate how different the LLM-generated translation text is from the pretraining data, we show the number of novel n -gram pairs that LLMs generated on WMT that have never been seen